UNIVERSITE CATHOLIQUE D'AFRIQUE CENTRALE
INSTITUT CATHOLIQUE DE YAOUNDE
Faculté de Sciences Sociales et de Gestion

A·O J FA

EXAMINATION OF DECISION THEORY

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BEM 3

Reflection questions (4 points)

1. After defining Risk, make a distinction between risk lover, risk averse and risk neutral individual
2. How are decisions chosen under the following criteria
   a) Maximax
   b) Minimax regret (savage criterion)
   c) Equal likelihood (Laplace)

Exercise 1 (9 points)

Mark and Sons company that develops inventions for the consumer market is launching two possible products that is the smoke fire detector and the motion detector. The smoke fire detector is presented as the best project as it detects flames as well as smoke. it will cost 100000frs and if it succeeds it will generate a revenue of 1000000frs. the motion detector which uses the conventional household lighting will cost 10000frs to develop and will generate 300000frs as revenue. However, analyst say that the smoke and fire detector have 50% chance of success while the motion detector has an 80% chance of success.

However, the smoke fire detectors must pass an underwriter's laboratories safety certification before it can be sold

* A commercial grade certification will result to 1000000frs sales but obtaining this certification is only 30%

* Residential grade certification is 60% likely, but would result in only 800000frs sales

* 10% chance that the smoke fire detector will not pass any certification test and in this case a complete failure for which the company will lose the initial 10000frs investment cost

Underwriters' laboratories charge a 5000frs non-refundable fee for the certification application.

a) Construct and evaluate the decision tree

b) Use the roll back method to determine the optimal decision

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Exercise 2 (7 points)

A manufacturing company wants to decide which commodity should stock to get maximum profit. It was supplied with the following information. The probability that the product will be in excess, normal and deficient is 0.50,0.20 and 0.30. The estimated profit or loss three commodities in respect of these different kinds of states are:

	Excess	Normal	Deficient
Yaourt	22000	-8400	36000
Fruit juice	8400	-6000	16000
Ice cream	8600	3000	-3200

Determine the optimal decision under each of the following decision criteria and show how you arrived at it:

(a) Maximax, (b) Maximin, (c) Minimax regret (savage criterion), (d) Equal likelihood (Laplace), (e) Hurwicz Alpha criterion α=0.7, (f)EMV, (g)EOL, (k)EVPI

Good luck!!!!